| Abbreviations |
| --- |
| ~M_4 = sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1) |
| ~M_5 = pk(sk1_1) |
| ~M_6 = hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))) |
| ~M_7 = sign(rpeQuote(mRPE,Good,hbind(pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1) |
| ~M_8 = pk(sk2_2) |
| ~M_9 = hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))) |
| ~X_1 = (P2,~M_7,~M_8,hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))<br>= (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1),pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))) |
| ~M_10 = sign(policyMsg(pk(sk2_2),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))),sk1_1) |
| ~X_2 = (P1,~M_4,~M_5,hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))<br>= (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))) |
| ~M_11 = sign(policyMsg(pk(sk1_1),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))),sk2_2) |
| ~M_12 = sign(confirmMsg(pk(sk2_2),hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),sk1_1) |
| ~M_13 = sign(confirmMsg(pk(sk1_1),hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))),sk2_2) |

A trace has been found.

Honest Process

Attacker

{1}new akRPE_1
{2}new akTEE_1
{3}new akADV_1

(~M,~M_1,~M_2,~M_3) = (pk(akRPE_1),pk(akTEE_1),pk(akADV_1),akADV_1)

1

Beginning of process rpe1

{9}new sk1_1

{11}event PolicyDeclared(P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

Beginning of process rpe2

(P1,~M_4,~M_5,~M_6)

PermissiveRpeTCB

{95}new sk2_2

{97}event PolicyDeclared(P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,~M_7,~M_8,~M_9)

~X_1

{20}event EvidenceChecked(P1,P2,pk(sk2_2),hpol(pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

{21}event QuotesComplete(P1)

{22}event PolicySent(P1,pk(sk2_2),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P1,P2,~M_10)

~X_2

{106}event EvidenceChecked(P2,P1,pk(sk1_1),hpol(pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

{107}event QuotesComplete(P2)

{108}event PolicySent(P2,pk(sk1_1),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

(P2,P1,~M_11)

(P2,P1,~M_11)

{26}event PolicyAuthenticated(P1,P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{29}event PolicyAccepted(P1,P2,pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{30}event PolicyAcceptedKeys(P1,P2,pk(sk1_1),pk(sk2_2),pol(P2,Members2,Both,Both,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{31}event PoliciesComplete(P1)

{34}event FullJoin(P1,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{35}event ConfirmationSent(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P1,P2,~M_12)

(P1,P2,~M_10)

{112}event PolicyAuthenticated(P2,P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{115}event PolicyAccepted(P2,P1,pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{116}event PolicyAcceptedKeys(P2,P1,pk(sk2_2),pk(sk1_1),pol(P1,Members2,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{117}event PoliciesComplete(P2)

{120}event FullJoin(P2,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))

{121}event ConfirmationSent(P2,P1,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

(P2,P1,~M_13)

(P2,P1,~M_13)

{39}event ConfirmationAuthenticated(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

{41}event ConfirmationAccepted(P1,P2,hstar(mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)))))

{42}event Confirm(P1,mkstar(OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2))))